

Reinforcement Learning-Based Cooperative Optimal Output Regulation via Distributed Adaptive Internal Model

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Abstract—In this article, a data-driven distributed control method is proposed to solve the cooperative optimal output regulation problem of leader–follower multiagent systems. Different from traditional studies on cooperative output regulation, a distributed adaptive internal model is originally developed, which includes a distributed internal model and a distributed observer to estimate the leader’s dynamics. Without relying on the dynamics of multiagent systems, we have proposed two reinforcement learning algorithms, policy iteration and value iteration, to learn the optimal controller through online input and state data, and estimated values of the leader’s state. By combining these methods, we have established a basis for connecting data-distributed control methods with adaptive dynamic programming approaches in general since these are the theoretical foundation from which they are built.

Index Terms—Adaptive optimal control, cooperative output regulation, distributed adaptive internal model, reinforcement learning.

I. INTRODUCTION

OVER the last two decades, the cooperative control of multiagent systems has drawn considerable attention due to their massive applications in many areas, including power system, unmanned aerial vehicles, wireless sensor networks, and distributed robotics [8], [29], [37]–[40]. Various cooperative control strategies have been proposed to address

consensus, flocking, and rendezvous problems in multiagent systems (see [11], [18], [22], [31], [33], [35], [51], and numerous references therein).

The celebrated output regulation theory considers the design of feedback controllers to achieve both disturbance rejection and asymptotic tracking for linear and nonlinear dynamical systems [19]. Two major strategies, feedback-feedforward control [14], [21] and internal model principle [9], [12], [45], have been leveraged for solving output regulation problems. Considering leader–follower multiagent systems [47]–[49] where the leader is modeled by an exosystem, cooperative output regulation has been a popular topic studied in the last decade [28], [30], [32], [41]. The extensions of cooperative output regulation can also be found in different control problems, such as event-triggered control [46] and fault-tolerant control [10].

A general assumption made in the literature of cooperative output regulation is that the leader’s dynamics is available to all the followers, which is usually not practical. Although one can transit the leader’s dynamics and state information to all the followers through communication channels, for multiagent systems with chain topology, such as a vehicular platoon, this centralized thinking is not scalable to large-scale networks of many autonomous agents due to time delays, package drops, and other network-induced effects. Before receiving the leader’s dynamics, each follower may opt to adopt its own open-loop controllers and observers, which may destabilize the whole multiagent systems. In order to remove this restrictive assumption, Cai *et al.* [6] and Huang [20] have developed distributed observers to estimate both leader’s dynamics and state information.

Optimality properties for the closed-loop multiagent systems are usually not discussed when solving cooperative output regulation problems. In order to fill in the gap between optimality and cooperative output regulation, our prior work [16] represents the first effort to combine adaptive optimal control, reinforcement learning [1]–[5], [7], [13], [15], [24], [26], [36], [44], [48], [51], and cooperative output regulation theories to develop distributed feedback-feedforward controllers that solve the cooperative optimal output regulation problem (CO²RP). By applying the optimal controller designed therein, each follower can achieve disturbance rejection and asymptotic tracking. Furthermore, the cost of the closed-loop system is minimized.

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As an extension, we have developed a data-driven solution to CO²RP in [17] incorporating a distributed internal model. Value iteration (VI) [1], [4] and policy iteration (PI) [14], [23] are two typically successive approximation approaches in reinforcement learning. When the system dynamics are not precisely known, a PI algorithm inspired by the main results mentioned in [14] for centralized systems is proposed therein to find successively approximations of the optimal controller. However, the implementation of PI algorithm usually relies on an initial stabilizing control policy. VI and a variant of adaptive dynamic programming methods [43] have been proposed to relax this strong assumption. However, these methods are usually studied for stabilization problems, not output regulation.

In this article, we present novel data-driven adaptive optimal control approaches to solve the CO²RP of leader-follower multiagent systems. Overall, this article has the following novel contributions.

- 1) This article is the first attempt to solve output regulation problems via VI. Compared with PI, a huge benefit of the VI approach is that no initial stabilizing control policy is required to start the learning process.
- 2) We have generalized our previous works [16], [17] on the CO²RP to a scenario where the leader's dynamics is only available to its successors. A straightforward application of distributed feedback-feedforward control or internal model principle cannot deal with this scenario since the leader's dynamics is not accessible to all followers. In order to achieve this generalization, we have developed a novel distributed adaptive internal model, which includes a distributed internal model and a distributed observer. Interestingly, the observer can estimate the leader's state and dynamics, which is used in the synthesis of our internal model. Essentially, this is a distributed version of the adaptive internal model [34].
- 3) During the learning phase, this article considers a more realistic case that followers cannot measure directly the leader's state for online learning. Instead, they can only measure the observation of the leader's state as a replacement. We have rigorously shown that the learned control gains will converge to any given small neighborhood of the optimal control gain.

The remainder of this technical paper is arranged as follows. In Section II, we formulate the cooperative output regulation problem of multiagent systems. A distributed adaptive internal model principle is introduced in Section III, and the CO²RP is also formulated therein. Data-driven adaptive optimal control design approaches are presented in Section IV. Simulation examples are demonstrated in Section V to verify the effectiveness of our proposed data-driven cooperative control methods. Finally, in Section VI, we give our conclusions.

Notations: Throughout this article, $\mathbb{C}^-$ stands for the open left-half complex plane. $\mathbb{R}$ denotes the set of real numbers, while $\mathbb{Z}_+$ denotes the set of nonnegative integers. $\mathcal{P}^n$ denotes the set of all $n \times n$ real, symmetric and positive semidefinite matrices. $|\cdot|$ represents the Euclidean norm for vectors and the induced norm for matrices. $\otimes$ indicates the Kronecker product operator and

$\text{vec}(A) = [a_1^T, a_2^T, \dots, a_m^T]^T$, where $a_i \in \mathbb{R}^n$ are the columns of $A \in \mathbb{R}^{n \times m}$. For a symmetric matrix $P \in \mathbb{R}^{m \times m}$, $\text{vecs}(P) = [p_{11}, 2p_{12}, \dots, 2p_{1m}, p_{22}, 2p_{23}, \dots, 2p_{m-1,m}, p_{mm}]^T \in \mathbb{R}^{(1/2)m(m+1)}$. For an arbitrary column vector $v \in \mathbb{R}^m$, $|v|_P$ denotes $v^T P v$. $\text{vecv}(v) = [v_1^2, v_1 v_2, \dots, v_1 v_m, v_2^2, v_2 v_3, \dots, v_{m-1} v_m, v_m^2]^T \in \mathbb{R}^{(1/2)m(m+1)}$.

II. PROBLEM FORMULATION AND PRELIMINARIES

Consider the following linear multiagent systems:

$$\begin{aligned} \dot{x}_i &= A_i x_i + B_i u_i + D_i v, \\ e_i &= C_i x_i + F v, \quad i = 1, 2, \dots, N \end{aligned} \quad (1)$$

where $x_i \in \mathbb{R}^{n_i}$, $u_i \in \mathbb{R}^{m_i}$, and $e_i \in \mathbb{R}^{p_i}$ are the state, control input, and tracking error of the i th subsystem, respectively, and $v \in \mathbb{R}^q$ is the state of an exosystem

$$\dot{v} = E v \quad (2)$$

which generates the disturbance $D_i v$ and the reference signal $y_0 = -F v$ to be tracked by the output $y_i = C_i x_i$ of each subsystem. Given the exosystem (2) and the plant (1), define a digraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. $\mathcal{V} = \{0, 1, \dots, N\}$ is the node set with node 0 denoting the leader modeled via the exosystem (2) and the remaining N nodes being identified as followers described by (1). $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$ refers to the edge set. The adjacency matrix $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{(N+1) \times (N+1)}$ is defined such that $a_{ij} > 0$ is a positive weight if $(j, i) \in \mathcal{E}$ and otherwise $a_{ij} = 0$. Then, the corresponding Laplacian $\mathcal{L}$ of $\mathcal{G}$ is

$$\mathcal{L} = \begin{bmatrix} \sum_{j=1}^N a_{0j} & -[a_{01}, \dots, a_{0N}] \\ -\Delta \mathbf{1}_N & \mathcal{H} \end{bmatrix}$$

where $\Delta = \text{diag}\{a_{10}, a_{20}, \dots, a_{N0}\}$, and $\mathcal{H} = [h_{ij}] \in \mathbb{R}^{N \times N}$ is defined by $h_{ii} = (\sum_{j=0}^N a_{ij}) - a_{ii}$ and $h_{ij} = -a_{ij}$ for all $i \neq j$. $\mathcal{N}_i$ defines a set of neighbors of the i th agent, and $\mathcal{N}_i^+ := \mathcal{N}_i \setminus \{0\}$.

Some standard assumptions for the solvability of traditional cooperative output regulation problem are made on the system (1) and (2).

Assumption 1: (A_i, B_i) is stabilizable, $\forall 1 \leq i \leq N$.

Assumption 2: The rank of the matrix $\begin{bmatrix} A_i - \lambda I & B_i \\ C_i & 0 \end{bmatrix}$ is $n_i + p_i$, $\forall \lambda \in \sigma(E)$, $\forall 1 \leq i \leq N$.

Assumption 3: The digraph $\mathcal{G}$ contains a directed spanning tree with the node 0 as the root.

Assumption 4: The state matrix E of the exosystem (leader) is a state matrix in the controllable canonical form (CCF). All the eigenvalues of E are simple with zero real part.

Assumption 5: E is only available to the leader and all of its successors.

Remark 1: Note that the Assumption 5 is a relaxation of our previous work [17] where the leader dynamics is available for all the agents. Assumption 4 is without loss of any generality. If a pair (E, b) is not in the CCF where $b = [0 \ 0 \ \dots \ 0 \ 1]^T$, the leader can easily find a transition matrix to convert E to a state matrix in the CCF.

The cooperative output regulation problem is solved if one designs a control policy such that the overall system is asymptotically stable (in the absence of v) and $\lim_{t \rightarrow \infty} e_i(t) = 0$,

for $i = 1, 2, \dots, N$. Note that the cooperative output regulation problem studied in this article includes leader–follower consensus problem as a special case.

The goal of this article is to solve the CO²RP that is an integration of optimal control and output regulation problems, with the ultimate objective to design a distributed control policy: 1) to solve the cooperative output regulation problem and 2) to minimize some predefined cost functions.

III. DISTRIBUTED ADAPTIVE INTERNAL MODEL

If the leader's dynamics E is available to all the followers, then one can develop a decentralized internal model as follows based on output regulation theory:

$$\dot{z}_i = Ez_i + G_2 e_i, \quad i = 1, 2, \dots, N. \quad (3)$$

where $G_2 = [0 \ 0 \ \dots \ 0 \ 1]^T$.

The implementation of decentralized internal model (3) depends on a condition that all the followers can directly communicate with the leader such that e_i is available. However, due to the communication constraints among agents, this condition is not realistic. In our previous work [17], we have developed a distributed internal model based on Assumption 3. Under a fixed digraph $\mathcal{G}$, the distributed internal model can be represented by

$$\dot{\hat{z}}_i = E\hat{z}_i + G_2 \hat{e}_i, \quad i = 1, 2, \dots, N \quad (4)$$

where the revised tracking error is

$$\hat{e}_i = \sum_{j \in \mathcal{N}_i} \frac{a_{ij}(y_i - y_j)}{\sum_{j=0}^N a_{ij}}.$$

However, for each follower, the construction of the distributed internal model still depends on the knowledge of dynamics E of exosystem (leader). Assumption 5 indicates that not all followers are fortunate to have that knowledge at the beginning, which renders that a straightforward application of traditional internal model principle [19] is not promising to overcome this challenge.

In this article, we propose an originally distributed adaptive internal model to compensate for the inaccessibility of the exosystem dynamics

$$\dot{E}_i = \mu_1 \sum_{j=0}^N a_{ij}(t)(E_j - E_i) \quad (5)$$

$$\dot{z}_i = E_i z_i + G_2 \hat{e}_i, \quad i = 1, 2, \dots, N \quad (6)$$

where $\mu_1 > 0$, and for the i th agent, $E_i = \mathbb{R}^{q \times q}$ is an estimation of exosystem dynamics E . For the leader, we simply have $\zeta_0 = v$ and $E_0 = E$.

Equation (5) forms an adaptive distributed observer of E for the multiagent systems (1). The asymptotic convergence of estimation errors of exosystem dynamics has been shown in [6, Lemma 2]. Then, it is checkable that, for all $i = 1, 2, \dots, N$, the pair $(\lim_{t \rightarrow \infty} E_i(t), G_2)$, or equivalently, (E, G_2) consolidates an internal model of the matrix E . Hence, (5)–(6) are distributed adaptive internal models for the system (1) and (2).

In the following theorem, we show that one can develop a state-feedback controller for the system (1) and (2) augmented with distributed adaptive internal model (5) and (6) to solve cooperative output regulation problem.

Theorem 1: Under Assumptions 1–5, the augmented system

$$\begin{aligned} \dot{v} &= Ev \\ \dot{x}_i &= A_i x_i + B_i u_i + D_i v \\ \dot{E}_i &= \mu_1 \sum_{j=0}^N a_{ij}(t)(E_j - E_i) \\ \dot{z}_i &= E_i z_i + G_2 \hat{e}_i \\ e_i &= C_i x_i + Fv, \quad i = 1, 2, \dots, N \end{aligned} \quad (7)$$

in closed-loop with the state-feedback controller

$$u_i = -K_{xi} x_i - K_{zi} z_i, \quad i = 1, 2, \dots, N \quad (8)$$

achieves cooperative output regulation if the matrix

$$A_{ci} = \begin{bmatrix} A_i - B_i K_{xi} & -B_i K_{zi} \\ G_2 C_i & E \end{bmatrix}$$

is Hurwitz.

Proof: Under Assumptions 1–5, for $i = 1, 2, \dots, N$, there exists a unique pair of matrices (X_i, U_i) solving the following regulator equations:

$$X_i E = A_i X_i + B_i U_i + D_i \quad (9)$$

$$0 = C_i X_i + F. \quad (10)$$

By the fact that A_{ci} is Hurwitz and [19, Lemma 1.27], the following matrix equations:

$$X_i E = (A_i - B_i K_{xi}) X_i - B_i K_{zi} Z_i + D_i, \quad (11)$$

$$Z_i E = E Z_i + G_2 (C_i X_i + F) \quad (12)$$

$$0 = C_i X_i + F$$

have a unique solution $(\hat{X}_i, Z_i)$.

Equation (11) is equivalent to

$$X_i E = A_i X_i + B_i (-K_{xi} X_i - K_{zi} Z_i) + D_i.$$

This implies that any pair (X_i, U_i) such that $X_i := \hat{X}_i$, and $U_i := -K_{xi} X_i - K_{zi} Z_i$ solve (9) and (10). Since the solution to (9) and (10) is unique, we can immediately obtain that the tuple (X_i, U_i, Z_i) is uniquely determined by (9)–(12).

For $i = 1, 2, \dots, N$, define $m_i = n_i + q$ and the following vectors and matrices:

$$\begin{aligned} \tilde{x}_i &= x_i - X_i v, \quad \tilde{z}_i = z_i - Z_i v, \quad \tilde{E}_i = E_i - E \\ \tilde{\zeta}_i &= \begin{bmatrix} \tilde{x}_i \\ \tilde{z}_i \end{bmatrix} \in \mathbb{R}^{m_i}, \quad \bar{C}_i = [C_i \ 0] \in \mathbb{R}^{1 \times m_i} \\ \bar{A}_i &= \begin{bmatrix} A_i & 0 \\ G_2 C_i & E \end{bmatrix}, \quad \bar{B}_i = \begin{bmatrix} B_i \\ 0 \end{bmatrix}, \quad K_i = [K_{xi} \ K_{zi}]. \end{aligned}$$

Based on (11) and (12), we have

$$\begin{aligned} \dot{\tilde{x}}_i &= A_i x_i + B_i u_i + D_i v - X_i E v \\ &= A_i x_i + B_i (-K_{xi} x_i - K_{zi} z_i) + D_i v - X_i E v \\ &= (A_i - B_i K_{xi}) x_i - B_i K_{zi} z_i + D_i v - X_i E v \\ &= (A_i - B_i K_{xi}) \tilde{x}_i - B_i K_{zi} \tilde{z}_i \end{aligned} \quad (13)$$

$$\begin{aligned}
\dot{\tilde{z}}_i &= E_i z_i + G_2 \hat{e}_i - Z_i E v \\
&= E z_i + \sum_{j \in \mathcal{N}_i} G_2 \frac{a_{ij}(y_i - y_j)}{\sum_{j=0}^N a_{ij}} - Z_i E v + \tilde{E}_i z_i \\
&= E z_i + G_2 C_i x_i - \sum_{j \in \mathcal{N}_i} G_2 \frac{a_{ij} C_j x_j}{\sum_{j=0}^N a_{ij}} - Z_i E v + \tilde{E}_i z_i \\
&= E \tilde{z}_i + G_2 C_i \tilde{x}_i - \sum_{j \in \mathcal{N}_i} G_2 \frac{a_{ij} C_j \tilde{x}_j}{\sum_{j=0}^N a_{ij}} + \tilde{E}_i z_i. \quad (14)
\end{aligned}$$

We can convert (13) and (14) to a more compact form

$$\begin{aligned}
\dot{\tilde{\xi}}_i &= A_{ci} \tilde{\xi}_i - \sum_{j \in \mathcal{N}_i} \frac{a_{ij}}{\sum_{j=0}^N a_{ij}} \begin{bmatrix} 0 \\ G_2 \tilde{C}_j \end{bmatrix} \tilde{\xi}_j + \begin{bmatrix} 0 & 0 \\ 0 & \tilde{E}_i \end{bmatrix} \tilde{\xi}_i + \begin{bmatrix} 0 \\ \tilde{E} v \end{bmatrix} \\
&:= A_{ci} \tilde{\xi}_i + L_{i0} \tilde{\xi}_{-i} + L_{i1}(t) \tilde{\xi}_i + L_{i2}(t) \\
e_i &= \tilde{C}_i \tilde{\xi}_i \quad (15)
\end{aligned}$$

where $\tilde{\xi}_{-i}$ is the vector lumping all the errors $\tilde{\xi}_1, \tilde{\xi}_2, \dots, \tilde{\xi}_N$ without $\tilde{\xi}_i$.

Based on Assumption 3, one can always label all the followers such that $i < j$ if $(i, j) \in \mathcal{E}$. Then, one can represent the overall multiagent systems by

$$\dot{\tilde{\xi}} = A_c \tilde{\xi} + L_1(t) \tilde{\xi} + L_2(t) \quad (16)$$

where

$$\begin{aligned}
\tilde{\xi} &= [\tilde{\xi}_1^T, \tilde{\xi}_2^T, \dots, \tilde{\xi}_N^T]^T \\
L_1 &= \text{blockdiag}(L_{11}, L_{21}, \dots, L_{N1}) \\
L_2 &= [L_{12}^T, L_{22}^T, \dots, L_{N2}^T]^T
\end{aligned}$$

and A_c is a block lower triangular matrix with submatrices A_{ci} on the diagonal, for any $i = 1, 2, \dots, N$.

Based on Assumption 4, the exostate $v(t)$ stays in a bounded set for any $t \geq 0$. The estimation error satisfies $\lim_{t \rightarrow \infty} \tilde{E}_i = 0$. Therefore, we have $\lim_{t \rightarrow \infty} L_1(t) = 0$, and $\lim_{t \rightarrow \infty} L_2(t) = 0$. Based on [6, Lemma 1], we have the origin of the system $\dot{\tilde{\xi}} = (A_c + L_1(t)) \tilde{\xi}$ is globally exponentially stable. This further implies that the system (16) is input-to-state stable regarding $L_2(t)$ as an external input. In other words, there exist a function β of class $\mathcal{KL}$ and a function γ of class $\mathcal{K}$ such that

$$|\tilde{\xi}(t)| \leq \beta(|\tilde{\xi}(t_0)|, t - t_0) + \gamma \left(\sup_{t_0 \leq \tau \leq t} |L_2(\tau)| \right). \quad (17)$$

Given any small $\epsilon > 0$, find $\epsilon_L > 0$ such that $\gamma(\epsilon_L) \leq \epsilon/2$. Since $\lim_{t \rightarrow \infty} L_2(t) = 0$, there always exists $T_1 > 0$ such that $|L_2(t)| \leq \epsilon_L$ for all $t \geq T_1$. Letting $t_0 \geq T_1$, we have

$$\begin{aligned}
|\tilde{\xi}(t)| &\leq \beta(|\tilde{\xi}(t_0)|, t - t_0) + \gamma(\epsilon_L) \\
&\leq \beta(c, t - t_0) + \frac{\epsilon}{2} \quad (18)
\end{aligned}$$

where $c > 0$ is the upper bound of $|\tilde{\xi}(t_0)|$. Since $\lim_{t \rightarrow \infty} \beta(c, t - t_0) = 0$, there always exists $T_2 \geq T_1$ such that $\beta(c, t - t_0) \leq \epsilon/2$, which implies that $|\tilde{\xi}(t)| \leq \epsilon$ for any $t \geq T_2$. One can conclude that $\lim_{t \rightarrow \infty} \tilde{\xi}_i(t) = 0$, and $\lim_{t \rightarrow \infty} e_i(t) = 0$ for each follower i . The proof is thus completed. $\square$

Remark 2: Note that, for $i = 1, 2, \dots, N$, the pair $(\bar{A}_i, \bar{B}_i)$ is stabilizable under Assumptions 1–5, which implies

that one can always find K_i such that $\sigma(A_{ci})$ is Hurwitz (see [19, Lemma 1.26]).

Remark 3: As (5) and (6) serve as a distributed adaptive internal model, the controller design in Theorem 1 inherits the advantages of internal models. First, the closed-loop system can tolerate certain plant uncertainty. Second, the designed control policy (8) and the distributed adaptive internal model rely on neither the measurement of exostate $v(t)$ nor its observation.

We define the following vectors and matrices to lump the states, inputs, and system matrices:

$$\begin{aligned}
\tilde{u} &= [\tilde{u}_1, \tilde{u}_2, \dots, \tilde{u}_N]^T \\
\bar{A} &= \text{blockdiag}(\bar{A}_1, \bar{A}_2, \dots, \bar{A}_N) \\
\bar{B} &= \text{blockdiag}(\bar{B}_1, \bar{B}_2, \dots, \bar{B}_N) \\
Q &= \text{blockdiag}(Q_1, Q_2, \dots, Q_N) \\
R &= \text{blockdiag}(R_1, R_2, \dots, R_N)
\end{aligned}$$

where, for $i = 1, 2, \dots, N$, $\tilde{u}_i = u_i - U_i v$, $Q_i = Q_i^T > 0$, and $R_i = R_i^T > 0$.

Now, we are ready to formulate CO²RP that will be solved in this article.

Problem 1 (CO²RP):

$$\min_{\tilde{u}} \int_0^\infty (\tilde{\xi}^T Q \tilde{\xi} + \tilde{u}^T R \tilde{u}) dt \quad (19)$$

$$\text{s.t. } \dot{\tilde{\xi}} = \bar{A} \tilde{\xi} + \bar{B} \tilde{u}. \quad (20)$$

Based on the linear optimal control theory, the optimal control policy solving Problem 1 is

$$\tilde{u}^* = -K^* \tilde{\xi}.$$

Since A, B, Q , and R are block diagonal matrices, we can derive that K^* is also a block diagonal matrix defined by $K^* = \text{blockdiag}(K_1^*, K_2^*, \dots, K_N^*)$, where the optimal control gain for the i th follower is

$$K_i^* = R_i^{-1} \bar{B}_i^T P_i^* := [K_{xi}^* \ K_{zi}^*]. \quad (21)$$

The matrix P_i^* is the unique solution to the following algebraic Riccati equation:

$$\bar{A}_i^T P_i^* + P_i^* \bar{A}_i + Q_i - P_i^* \bar{B}_i R_i^{-1} \bar{B}_i^T P_i^* = 0. \quad (22)$$

Therefore, the optimal controller is equivalent to

$$\begin{aligned}
u_i^* &= \tilde{u}_i^* + U_i v \\
&= -K_i^* \tilde{\xi}_i + U_i v \\
&= -K_{xi}^* \tilde{x}_i - K_{zi}^* \tilde{z}_i + U_i v \\
&= -K_{xi}^* x_i - K_{zi}^* z_i, \quad i = 1, 2, \dots, N \quad (23)
\end{aligned}$$

which is in a similar form with the controller (8).

Moreover, since the pair $(\bar{A}_i, \bar{B}_i)$ is stabilizable, it is straightforward to have the system matrix of closed-loop system

$$\begin{bmatrix} A_i - B_i K_{xi}^* & -B_i K_{zi}^* \\ G_2 C_i & E \end{bmatrix}$$

is Hurwitz for all the followers. Based on Theorem 1, the designed optimal controller (23) can be used to solve the cooperative output regulation problem.

IV. ADAPTIVE OPTIMAL CONTROLLER DESIGN

The key point to solve Problem 1 is to solve the Riccati equation (22), which usually depends on the knowledge of system matrices. In this section, we will design two data-driven adaptive optimal control policies, PI and VI, to solve the CO²RP. The developed approaches are able to approximate the control gains K_i^* for each follower without relying on the knowledge of system matrices A_i , B_i , and D_i .

In both PI and VI designs, the leader's state information is required by all followers. To accommodate this requirement, we develop the following estimator of leader's state v :

$$\dot{\zeta}_i = E_i \zeta_i + \mu_2 \sum_{j \in \mathcal{N}_i^+} a_{ij} (\zeta_j - \zeta_i) \quad (24)$$

where $\mu_2 > 0$. It has been shown in [6] that, for $i = 1, 2, \dots, N$, the estimation error $\tilde{\zeta}_i := \zeta_i - v$ converges to zero as the time goes to infinity.

A. Online PI Design

To begin with, we rewrite the augmented system (7) by

$$\begin{aligned} \dot{\zeta}_i &= \bar{A}_i \zeta_i + \bar{B}_i u_i + \begin{bmatrix} D_i \\ \alpha_{i0} G_2 F \end{bmatrix} v - \sum_{j \in \mathcal{N}_i^+} \alpha_{ij} \begin{bmatrix} 0 \\ G_2 \bar{C}_j \end{bmatrix} \zeta_j \\ &\quad + \begin{bmatrix} 0 \\ \tilde{E}_i z_i \end{bmatrix} \\ &= \bar{A}_i^{(k)} \zeta_i + \bar{B}_i (K_i^{(k)} \zeta_i + u_i) + \begin{bmatrix} D_i \\ \alpha_{i0} G_2 F \end{bmatrix} \zeta_i \\ &\quad - \sum_{j \in \mathcal{N}_i^+} \alpha_{ij} \begin{bmatrix} 0 \\ G_2 \bar{C}_j \end{bmatrix} \zeta_j + \begin{bmatrix} -D_i \tilde{\zeta}_i \\ \tilde{E}_i z_i + \alpha_{i0} G_2 F \tilde{\zeta}_i \end{bmatrix} \end{aligned}$$

where, for $i = 1, 2, \dots, N$, $\bar{A}_i^{(k)} = \bar{A}_i - \bar{B}_i K_i^{(k)}$, and $\alpha_{ij} = a_{ij} / (\sum_{j=0}^N a_{ij})$.

By increasing parameters μ_1 and μ_2 , one can accelerate the convergence of estimation errors $\tilde{E}_i$ and $\tilde{\zeta}_i$. Thus, the term

$$\rho_i(t) := \begin{bmatrix} -D_i \tilde{\zeta}_i \\ \tilde{E}_i z_i + \alpha_{i0} G_2 F \tilde{\zeta}_i \end{bmatrix}$$

will converge to zero with any desired speed by tuning μ_1 and μ_2 .

The idea of online PI is to implement both policy evaluation

$$\begin{aligned} 0 &= (\bar{A}_i - \bar{B}_i K_i^{(k)})^T P_i^{(k)} + P_i^{(k)} (\bar{A}_i - \bar{B}_i K_i^{(k)}) \\ &\quad + Q_i + (K_i^{(k)})^T R_i K_i^{(k)} \quad (25) \end{aligned}$$

and policy improvement

$$K_i^{(k+1)} = R_i^{-1} \bar{B}_i^T P_i^{(k)} \quad (26)$$

using online data by iterations.

By (25), letting $\rho_i = 0$, we have

$$\begin{aligned} &|\zeta_i(t + \delta t)|_{P_i^{(k)}} - |\zeta_i(t)|_{P_i^{(k)}} \\ &= \int_t^{t+\delta t} \left[|\zeta_i|_{(\bar{A}_i^{(k)})^T P_i^{(k)} + P_i^{(k)} \bar{A}_i^{(k)}} + 2\zeta_i^T \Theta_{i0}^T P_i^{(k)} \zeta_i \right. \\ &\quad \left. + 2(u_i + K_i^{(k)} \zeta_i)^T \bar{B}_i^T P_i^{(k)} \zeta_i - 2 \sum_{j \in \mathcal{N}_i^+} \zeta_j^T \Theta_{ij}^T P_i^{(k)} \zeta_i \right] d\tau \end{aligned}$$

$$\begin{aligned} &= \int_t^{t+\delta t} \left[-|\zeta_i|_{Q_i + (K_i^{(k)})^T R_i K_i^{(k)}} + 2\zeta_i^T \Theta_{i0}^T P_i^{(k)} \zeta_i + 2(u_i + K_i^{(k)} \zeta_i)^T \right. \\ &\quad \left. \times R_i K_i^{(k+1)} \zeta_i - 2 \sum_{j \in \mathcal{N}_i^+} \zeta_j^T \Theta_{ij}^T P_i^{(k)} \zeta_i \right] d\tau \quad (27) \end{aligned}$$

where, for $j = j_1, j_2, \dots, j_{|\mathcal{N}_i^+|} \in \mathcal{N}_i^+$, $\Theta_{ij} = \begin{bmatrix} 0 \\ \alpha_{ij} G_2 \bar{C}_j \end{bmatrix}$ and $\Theta_{i0} = \begin{bmatrix} D_i \\ \alpha_{i0} G_2 F \end{bmatrix}$.

Letting

$$\begin{aligned} \psi_i &= \begin{bmatrix} \zeta_{j_1}^T, \zeta_{j_2}^T, \dots, \zeta_{j_{|\mathcal{N}_i^+|}}^T \end{bmatrix}^T \in \mathbb{R}^{p_i} \\ \Xi_i^{(k)} &= \begin{bmatrix} P_i^{(k)} \Theta_{ij_1}, P_i^{(k)} \Theta_{ij_2}, \dots, P_i^{(k)} \Theta_{ij_{|\mathcal{N}_i^+|}} \end{bmatrix}^T \in \mathbb{R}^{p_i \times m_i} \quad (28) \end{aligned}$$

by the Kronecker product representation, we obtain

$$\begin{aligned} &|\zeta_i|_{Q_i + (K_i^{(k)})^T R_i K_i^{(k)}} \\ &= (\zeta_i^T \otimes \zeta_i^T) \text{vec} \left(Q_i + (K_i^{(k)})^T R_i K_i^{(k)} \right) \\ &\quad \zeta_i^T \Theta_{i0}^T P_i^{(k)} \zeta_i \\ &= (\zeta_i^T \otimes \zeta_i^T) \text{vec} \left(\Theta_{i0}^T P_i^{(k)} \right) \\ &\quad \sum_{j \in \mathcal{N}_i^+} \zeta_j^T \Theta_{ij}^T P_i^{(k)} \zeta_i \\ &= (\zeta_i^T \otimes \psi_i^T) \text{vec} \left(\Xi_i^{(k)} \right) \\ &\quad (u_i + K_i^{(k)} \zeta_i)^T R_i K_i^{(k+1)} \zeta_i \\ &= \left[(\zeta_i^T \otimes \zeta_i^T) \left(I \otimes (K_i^{(k)})^T R_i \right) + (\zeta_i^T \otimes u_i^T) (I \otimes R_i) \right] \\ &\quad \times \text{vec} \left(K_i^{(k+1)} \right). \end{aligned}$$

Moreover, for any two vectors a, b and a sufficiently large number $s > 0$, define

$$\begin{aligned} \delta_a &= [\text{vecv}(a(t_1)) - \text{vecv}(a(t_0)), \dots, \\ &\quad \text{vecv}(a(t_s)) - \text{vecv}(a(t_{s-1}))]^T \\ \Gamma_{a,b} &= \left[\int_{t_0}^{t_1} a \otimes b d\tau, \int_{t_1}^{t_2} a \otimes b d\tau, \dots, \int_{t_{s-1}}^{t_s} a \otimes b d\tau \right]^T. \end{aligned}$$

Equation (27) implies the following linear equation:

$$\Psi_{\text{PI},i}^{(k)} \begin{bmatrix} \text{vecs}(P_i^{(k)}) \\ \text{vec}(K_i^{(k+1)}) \\ \text{vec}(\Theta_{i0}^T P_i^{(k)}) \\ \text{vec}(\Xi_i^{(k)}) \end{bmatrix} = \Phi_{\text{PI},i}^{(k)} \quad (29)$$

where

$$\begin{aligned} \Psi_{\text{PI},i}^{(k)} &= \left[\delta_{\zeta_i}, -2\Gamma_{\zeta_i \zeta_i} \left(I \otimes (K_i^{(k)})^T R_i \right) - 2\Gamma_{\zeta_i u_i} (I \otimes R_i) \right. \\ &\quad \left. - 2\Gamma_{\zeta_i \zeta_i}, 2\Gamma_{\zeta_i \psi_i} \right] \\ \Phi_{\text{PI},i}^{(k)} &= -\Gamma_{\zeta_i \zeta_i} \text{vec} \left(Q_i + (K_i^{(k)})^T R_i K_i^{(k)} \right). \end{aligned}$$

The uniqueness of solution to (29) is guaranteed under some rank condition as shown next. For want of space, we omit the proof of lemma which follows the same line of proofs as in [14] and [23].

Lemma 1: For all $k \in \mathbb{Z}_+$, if there exists $s^* \in \mathbb{Z}_+$ such that for all $s > s^*$

$$\text{rank}([\Gamma_{\xi_i \xi_i}, \Gamma_{\xi_i u_i}, \Gamma_{\xi_i \xi_i}, \Gamma_{\xi_i, \psi_i}]) = \frac{m_i^2}{2} + \left(\frac{3}{2} + q + p_i\right)m_i \quad (30)$$

then the matrix $\Psi_{\text{PI},i}^{(k)}$ has full column rank for all $k \in \mathbb{Z}_+$.

Now, we are ready to present a data-driven PI Algorithm 1 to approximate the optimal values K_i^* and P_i^* .

Algorithm 1 Online PI Algorithm for CO²RP

- 1: Select a small $c_1 > 0$. Apply $u_i = -K_i^{(0)}\xi_i + v_i$ on $[t_0, t_s]$ with v_i an exploration noise, s.t. (30) holds $\forall 1 \leq i \leq N$.
- 2: $i \leftarrow 1$
- 3: **repeat**
- 4: $k \leftarrow -1$
- 5: **repeat**
- 6: $k \leftarrow k + 1$
- 7: Solve $P_i^{(k)}$ and $K_i^{(k+1)}$ from (29)
- 8: **until** $|P_i^{(k)} - P_i^{(k-1)}| < c_1$ for $k \geq 1$
- 9: $P_i^* \leftarrow P_i^{(k)}$
- 10: The learned control policy is

$$u_i = -K_i^{(k+1)}\xi_i := -K_i^*\xi_i \quad (31)$$

- 11: $i \leftarrow i + 1$
 - 12: **until** $i = N + 1$
-

B. Online VI Design

Based on [4], the nature of VI algorithm is to update the value matrix and control gain by

$$\begin{aligned} P_i^{(k+1)} &= \epsilon_k (\bar{A}_i^T P_i^{(k)} + P_i^{(k)} \bar{A}_i - P_i^{(k)} \bar{B}_i R_i^{-1} B_i^T P_i^{(k)} + Q_i) \\ &\quad + P_i^{(k)} \\ K_i^{(k)} &= R_i^{-1} \bar{B}_i^T P_i^{(k)} \end{aligned} \quad (32)$$

where ϵ_k is the step size satisfying

$$\epsilon_k > 0, \quad \sum_{k=0}^{\infty} \epsilon_k = \infty, \quad \sum_{k=0}^{\infty} \epsilon_k^2 < \infty. \quad (33)$$

The online VI design tries to update $P_i^{(k)}$ avoid using the knowledge of $\bar{A}_i$ and $\bar{B}_i$. Based on (32), we take the derivative of a function $\xi_i^T P_i^{(k)} \xi_i$, i.e.,

$$\begin{aligned} &\frac{d}{dt}(\xi_i^T P_i^{(k)} \xi_i) \\ &= \left(\bar{A}_i \xi_i + \bar{B}_i u_i + \Theta_{i0} \xi_i - \sum_{j \in \mathcal{N}_i^+} a_{ij} \Theta_{ij} \xi_j + \rho_i \right)^T P_i^{(k)} \xi_i \\ &\quad + \xi_i^T P_i^{(k)} \left(\bar{A}_i \xi_i + \bar{B}_i u_i + \Theta_{i0} \xi_i - \sum_{j \in \mathcal{N}_i^+} a_{ij} \Theta_{ij} \xi_j + \rho_i \right) \end{aligned}$$

$$\begin{aligned} &= \xi_i^T \mathcal{H}_i^{(k)} \xi_i + 2(R_i u_i)^T K_i^{(k)} \xi_i + 2\xi_i^T \Theta_{i0}^T P_i^{(k)} \xi_i \\ &\quad - 2 \sum_{j \in \mathcal{N}_i^+} \xi_j^T \Theta_{ij}^T P_i^{(k)} \xi_i + 2\rho_i^T P_i^{(k)} \xi_i \end{aligned} \quad (34)$$

where $\mathcal{H}_i^{(k)} = \bar{A}_i^T P_i^{(k)} + P_i^{(k)} \bar{A}_i$.

Letting $\rho_i = 0$, (34) implies the following equation:

$$\Psi_{\text{VI},i} \begin{bmatrix} \text{vecs}(\mathcal{H}_i^{(k)}) \\ \text{vec}(K_i^{(k)}) \\ \text{vec}(\Theta_{i0}^T P_i^{(k)}) \\ \text{vec}(\Xi_i^{(k)}) \end{bmatrix} = \Phi_{\text{VI},i}^{(k)} \quad (35)$$

where

$$\begin{aligned} \Psi_{\text{VI},i} &= [\delta_{\xi_i}, 2\Gamma_{\xi_i u_i}(I \otimes R_i), 2\Gamma_{\xi_i \xi_i}, -2\Gamma_{\xi_i, \psi_i}] \\ \Phi_{\text{VI},i}^{(k)} &= -\Gamma_{\xi_i \xi_i} \text{vec}(P_i^{(k)}). \end{aligned}$$

Similar to PI Algorithm 1, under the condition of Lemma 1, (35) has a unique least-squares solution.

By defining a collection of bounded sets $\{\mathcal{B}_q\}_{q=0}^{\infty}$ as follows:

$$\mathcal{B}_q \subset \mathcal{B}_{q+1}, q \in \mathbb{Z}_+, \quad \lim_{q \rightarrow \infty} \mathcal{B}_q = \mathcal{P}_n. \quad (36)$$

The online VI algorithm is given in Algorithm 2.

Algorithm 2 Online VI Algorithm for CO²RP

- 1: Select a small $c_2 > 0$. Apply any locally essentially bounded input u_i on $[t_0, t_s]$ s.t. (30) holds $\forall 1 \leq i \leq N$.
 - 2: $i \leftarrow 1$.
 - 3: **repeat**
 - 4: $k \leftarrow 0, q \leftarrow 0$. Choose a symmetric $P_i^{(0)} > 0$.
 - 5: **loop**
 - 6: Solve $\mathcal{H}_i^{(k)}$ and $K_i^{(k)}$ from (35)
 - 7: $\tilde{P}_i^{(k+1)} \leftarrow P_i^{(k)} + \epsilon_k(\mathcal{H}_i^{(k)} + Q_i - (K_i^{(k)})^T R_i K_i^{(k)})$
 - 8: **if** $\tilde{P}_i^{(k+1)} \notin \mathcal{B}_q$ **then**
 - 9: $P_i^{(k+1)} \leftarrow P_i^{(0)}, q \leftarrow q + 1$.
 - 10: **else if** $|P_i^{(k)} - P_i^{(k-1)}|/\epsilon_k < c_2$ **then**
 - 11: **return** $(P_i^{(k)}, K_i^{(k)})$ as (P_i^*, K_i^*)
 - 12: **else** $P_i^{(k+1)} \leftarrow \tilde{P}_i^{(k+1)}$
 - 13: **end if**
 - 14: $k \leftarrow k + 1$
 - 15: **end loop**
 - 16: $i \leftarrow i + 1$
 - 17: **until** $i = N + 1$
-

Remark 4: Like traditional adaptive control [50] and existing work on ADP [23], [27], we add an exploration noise to the input during the learning phase in order to satisfy (30). The proposed Algorithms 1 and 2 are off-policy methods [42], where the introduction of exploration noise does not affect the accuracy of the learned control gains at each iteration.

C. Convergence Analysis

Before showing Algorithms 1 and 2, the following lemmas are introduced to discuss the relationship between the control

gain and value solved by online algorithms and that solved by traditional model-based approaches.

Lemma 2: Let sequences $\{\bar{P}_i^{(k)}\}_{k=1}^{\infty}$ and $\{\bar{K}_i^{(k)}\}_{k=1}^{\infty}$ be the solution obtained from model-based PI approaches (25) and (26). If (30) is satisfied, then, for any small $c_3 > 0$, there exist $\mu_1, \mu_2 > 0$ such that, for $i = 1, 2, \dots, N$, $\{P_i^{(k)}\}$ and $\{K_i^{(k)}\}$ computed by Algorithm 1 at any iteration k satisfying the inequalities $|P_i^{(k)} - \bar{P}_i^{(k)}| < c_3$ and $|K_i^{(k)} - \bar{K}_i^{(k)}| < c_3$.

Proof: See the Appendix. $\square$

Lemma 3: Let sequences $\{\hat{P}_i^{(k)}\}_{k=1}^{\infty}$ and $\{\hat{K}_i^{(k)}\}_{k=1}^{\infty}$ be the solution obtained from model-based VI approaches (32). If (30) is satisfied, then, for any small $c_3 > 0$, there exist $\mu_1, \mu_2 > 0$ such that, for $i = 1, 2, \dots, N$, $\{P_i^{(k)}\}$ and $\{K_i^{(k)}\}$ computed by Algorithm 2 at any iteration k satisfying the inequalities $|P_i^{(k)} - \hat{P}_i^{(k)}| < c_3$ and $|K_i^{(k)} - \hat{K}_i^{(k)}| < c_3$.

Proof: See the Appendix. $\square$

The convergences of Algorithms 1 and 2 are analyzed in Theorem 2.

Theorem 2: If (30) is satisfied, then, for any small $c_4 > 0$, there exist $\mu_1, \mu_2 > 0$ and $k^* \in \mathbb{Z}_+$ such that, for $i = 1, 2, \dots, N$, sequences $\{P_i^{(k)}\}_{k=1}^{\infty}$ and $\{K_i^{(k)}\}_{k=1}^{\infty}$ computed by Algorithms 1 and 2 satisfying the inequalities $|P_i^{(k^*)} - P_i^*| < c_4$ and $|K_i^{(k^*)} - K_i^*| < c_4$.

Proof: The convergence of model-based PI approach has been shown in [25], while that of model-based VI has been shown recently in [4]. For each follower i , it is direct to have that there exists k^* such that

$$\begin{aligned} |\bar{K}_i^{(k^*)} - K_i^*| &< c_4/2 \\ |\bar{P}_i^{(k^*)} - P_i^*| &< c_4/2 \\ |\hat{K}_i^{(k^*)} - K_i^*| &< c_4/2 \\ |\hat{P}_i^{(k^*)} - P_i^*| &< c_4/2. \end{aligned}$$

Furthermore, by Lemmas 2 and 3, there exists μ_1 and μ_2 such that

$$\begin{aligned} |K_i^{(k^*)} - \bar{K}_i^{(k^*)}| &< c_4/2 \\ |P_i^{(k^*)} - \bar{P}_i^{(k^*)}| &< c_4/2 \\ |K_i^{(k^*)} - \hat{K}_i^{(k^*)}| &< c_4/2 \\ |P_i^{(k^*)} - \hat{P}_i^{(k^*)}| &< c_4/2. \end{aligned}$$

The proof is completed based on triangle inequality. $\square$

V. ILLUSTRATIVE EXAMPLES

In this section, we validate the proposed learning algorithms by two examples.

A. Example 1

We first consider an example of four followers to demonstrate the obtained theoretical results. For $i = 1, 2, 3, 4$, the system matrices are

$$\begin{aligned} A_i &= \begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B_i = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad D_i = \begin{bmatrix} 0 & 0 \\ 0 & 1.5 \times i \end{bmatrix} \\ C_i &= \begin{bmatrix} 1 \\ 0 \end{bmatrix}^T, \quad F = \begin{bmatrix} -1 \\ 0 \end{bmatrix}^T, \quad E = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}. \end{aligned}$$

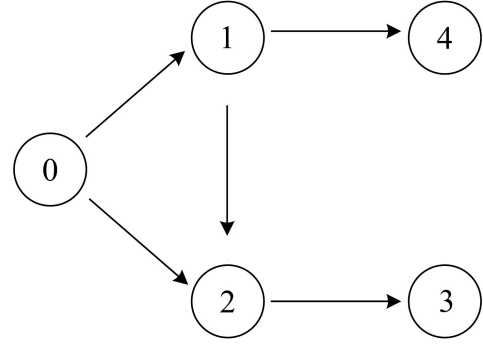


Fig. 1. Communication graph of Example 1 (Agent #0 is the leader, and agents #1 – 4 are followers).

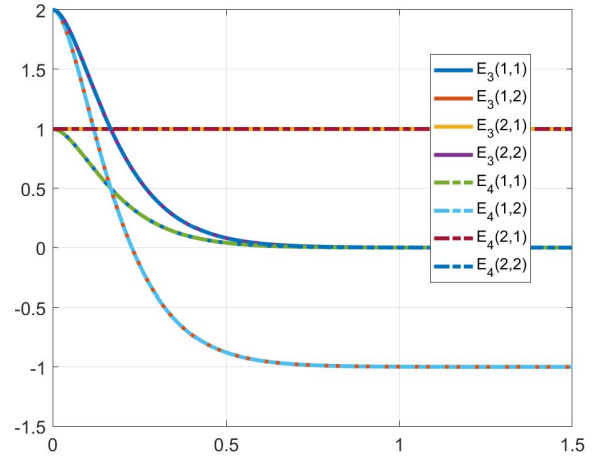


Fig. 2. Estimation of exosystem dynamics for agents #3 and 4 in Example 1.

For $i = 1, 2, 3, 4$, the weight matrices Q_i and R_i are identity matrices. The communication graph of the multiagent system is shown in Fig. 1. Suppose that the system matrices A_i , B_i , and D_i are unknown and E is only known to leader and its successors. The observer (5) has been designed to observe exosystem dynamics E , where the estimation by followers #3, #4 has been shown in Fig. 2. One can check that the estimation has achieved convergence when $t = 1$ s. We have collected the input and state data from $t = 5$ s to $t = 15$ s for learning the optimal control gains. Then, we leverage these online data to learn the optimal value and optimal control gains through Algorithms 1 and 2.

Fig. 3 shows that the value of $P_i^{(k)}$ for the k th iteration of the i th follower and its comparison of the optimal value P_i^* under PI Algorithm 1. All the agents can finish the learning in less than 16 iterations. The control gains learned by PI are

$$\begin{aligned} K_1^{(12)} &= [1.1214 \ 3.0598 \ -1.1813 \ 0.7775] \\ K_2^{(16)} &= [1.1231 \ 3.0585 \ -1.1816 \ 0.7778] \\ K_3^{(12)} &= [1.1217 \ 3.0597 \ -1.1814 \ 0.7776] \\ K_4^{(8)} &= [1.1010 \ 3.0497 \ -1.1686 \ 0.7667]. \end{aligned} \quad (37)$$

For comparison, the optimal control gains are computed as follows:

$$K_i^* = [1.1215 \ 3.0598 \ -1.1813 \ 0.7775], \quad i = 1, 2, 3, 4. \quad (38)$$

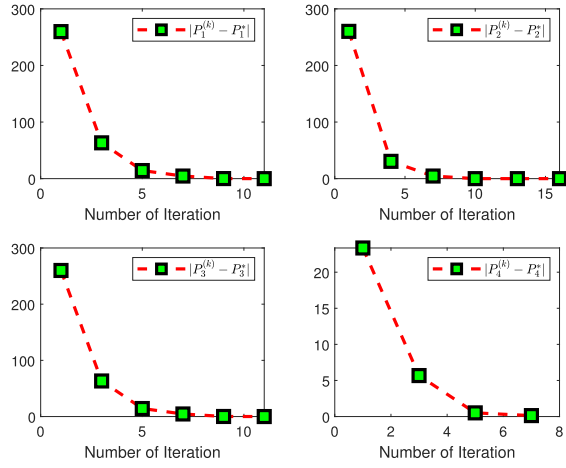


Fig. 3. Comparisons of $P_i^{(k)}$ and P_i^* of each follower under PI Algorithm 1 in Example 1.

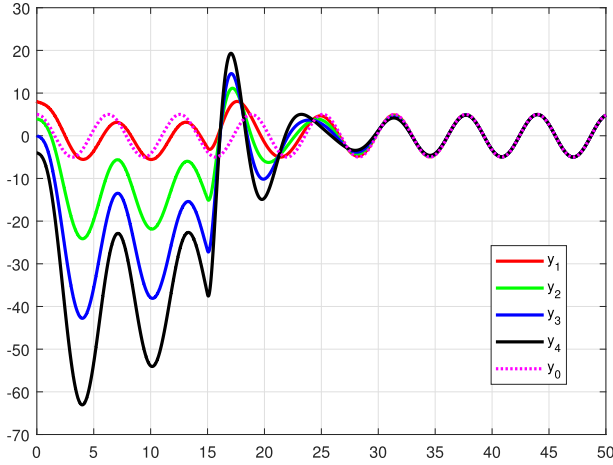


Fig. 4. Evolution of the system outputs y_i and reference signal y_0 under PI Algorithm 1 in Example 1.

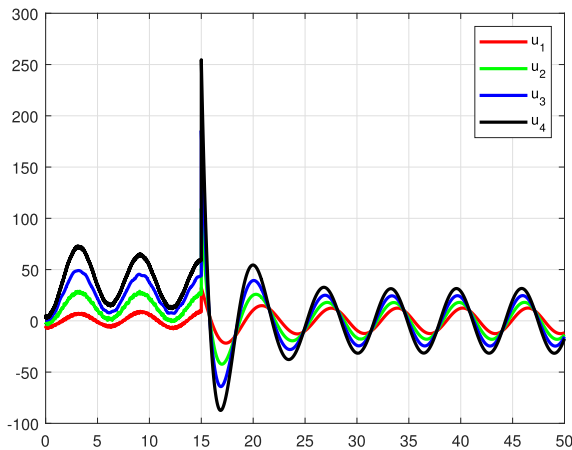


Fig. 5. Evolution of distributed control inputs under PI Algorithm 1 in Example 1.

One can observe that the learned control gains are close to their optimal ones. Under PI Algorithm 1, the system outputs of all leader and follower are described in Fig. 4, and Fig. 5 shows distributed controller inputs.

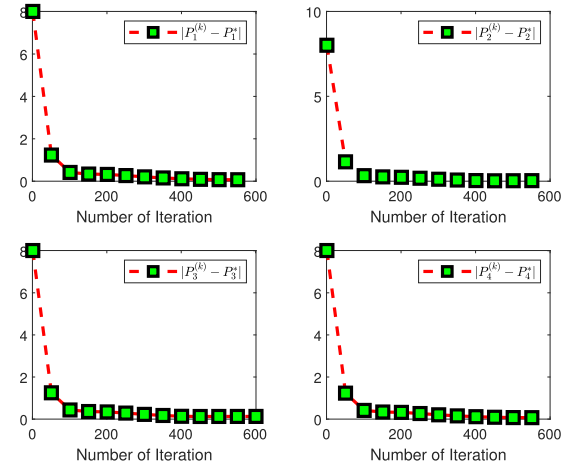


Fig. 6. Comparisons of $P_i^{(k)}$ and P_i^* of each follower under VI Algorithm 2 in Example 1.

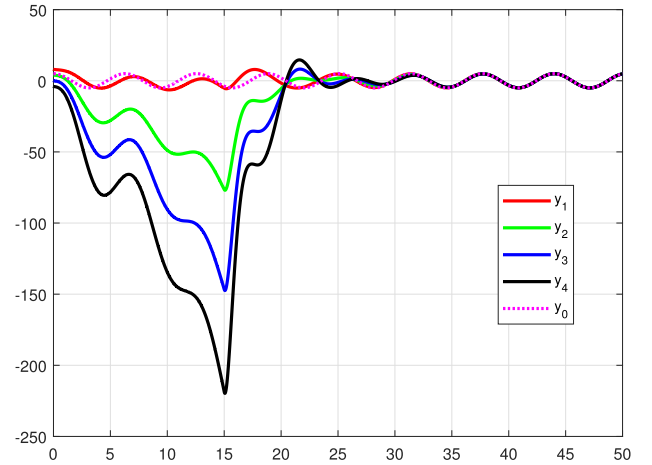


Fig. 7. Evolution of the system outputs y_i and reference signal y_0 under VI Algorithm 2 in Example 1.

The comparison of $P_i^{(k)}$ under VI Algorithm 2 can be found in Fig. 6. The learned control gains by VI are

$$\begin{aligned} K_1^{(585)} &= [1.1156 \ 3.0522 \ -1.1722 \ 0.7748] \\ K_2^{(576)} &= [1.1291 \ 3.0531 \ -1.1842 \ 0.7797] \\ K_3^{(612)} &= [1.0766 \ 3.0864 \ -1.1628 \ 0.7677] \\ K_4^{(545)} &= [1.1616 \ 3.0752 \ -1.2405 \ 0.7962]. \end{aligned} \quad (39)$$

We see that VI algorithms need over 500 iterations, which are much more than PI to converge. However, the VI algorithm is implementable even if the initial control policy is not admissible.

The system outputs and their reference are shown in Fig. 7. It can be checked that initial control exerted from $t = 0$ s to $t = 15$ s is not stabilizing. Interestingly, one can still learn an approximated optimal and stabilizing control policy using VI Algorithm 2.

B. Example 2

In the second example, we consider a platoon of four connected and autonomous vehicles modeled

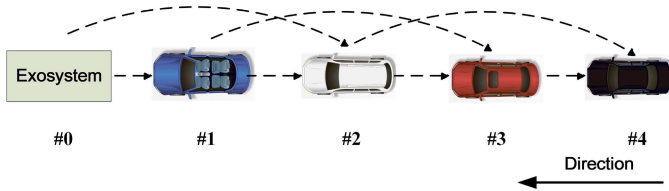


Fig. 8. Structure of a vehicular platoon in Example 2.

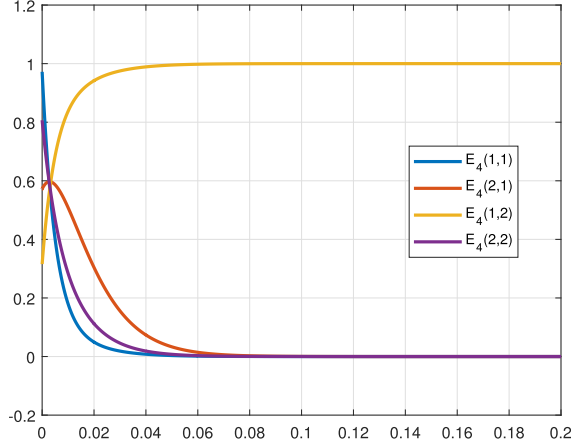
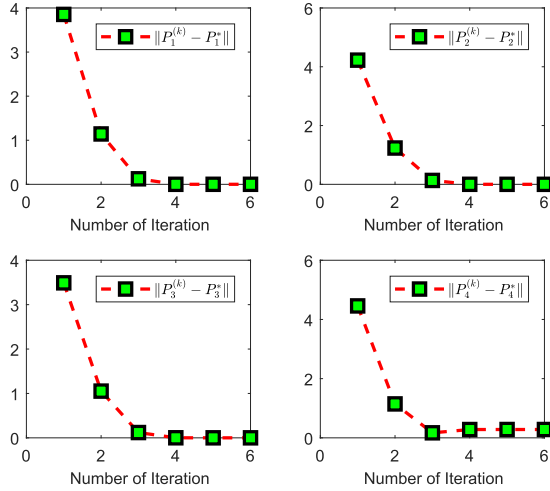


Fig. 9. Estimation of exosystem dynamics for the fourth vehicle in Example 2.

Fig. 10. Comparisons of $P_i^{(k)}$ and P_i^* of each vehicle under PI Algorithm 1 in Example 2.

in [16]

$$\begin{aligned} \dot{p}_i &= v_i \\ \dot{v}_i &= a_i \\ \dot{a}_i &= \tau_i^{-1} a_i + \tau_i^{-1} u_i + d_i \end{aligned} \quad (40)$$

where p_i , v_i , a_i , and τ_i are the position, velocity, acceleration, and time constant of the engine of vehicle $\#i$, respectively. The constant d_i is the ratio of the mechanical drag to the product of τ_i and the mass of vehicle $\#i$. For $i = 1, 2, 3, 4$, the values of τ_i and d_i are the same as that in [16]. The topology of this platoon is shown in Fig. 8. Both the desired position p_i^* and

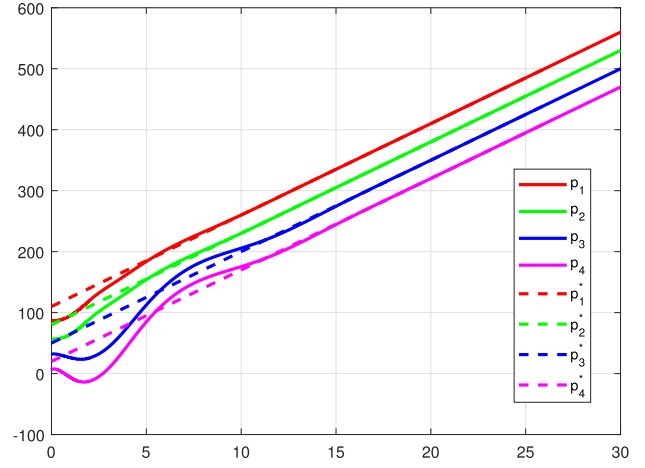
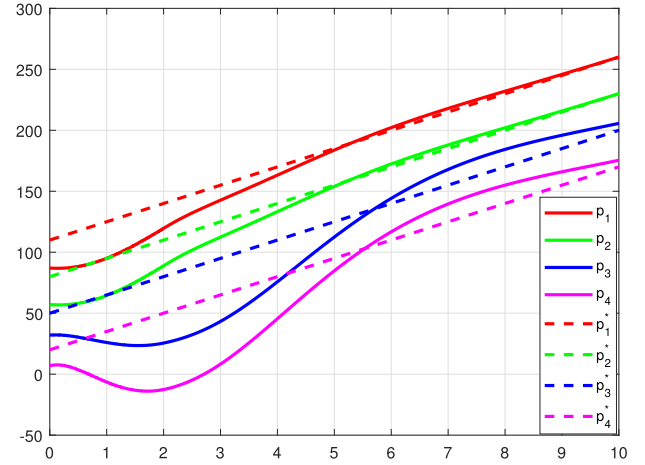
Fig. 11. Actual trajectories (p_i) of connected and autonomous vehicles and their references (p_i^*).

Fig. 12. Zoomed trajectories of connected and autonomous vehicles.

disturbance d_i are generated by the following exosystem:

$$\begin{aligned} \dot{v} &= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} v \\ d_i &= d_i v_2 \\ p_i^* &= -15v_1 - 20v_2 + 30(4-i) \end{aligned} \quad (41)$$

with the initial value $v = [0 \ 1]^T$. In this setting, vehicles will maintain 30 m as a safe headway at the steady state. From $t = 0$ s to $t = 1$ s, we let the estimation of exosystem dynamics and exostate converge. The estimation of E in vehicle $\#4$ is shown in Fig. 9. We collect the data from $t = 1$ s to $t = 2$ s for PI learning. The convergence of P_i for each vehicle is shown in Fig. 10. It is shown that all the vehicles can learn the optimal control values after six iterations. We update the control policy and the trajectories of vehicles can be checked in Fig. 11 and the zoomed of Fig. 12 where all the vehicles are able to track their references after 10 s.

VI. CONCLUSION

In this article, a distributed adaptive optimal framework has been proposed for solving the leader–following cooperative

output regulation problem of multiagent systems using the internal model principle and reinforcement learning techniques. By minimizing a predefined quadratic cost function, we have developed a computational data-driven design method for adaptive optimal control of multiagent systems. In the absence of the precise knowledge of the system dynamics, desirable controllers are learned from data through two reinforcement learning strategies: PI and VI. Two numerical examples are presented to demonstrate the feasibility of the presented learning-based cooperative output regulation methods. Our future work will concentrate on the extension to nonlinear CO²RP by overcoming technical challenges, such as constructing a distributed adaptive internal model for nonlinear systems, and approximating the nonlinear optimal control policy.

APPENDIX A PROOF OF LEMMA 2

In order to complete the proof of Lemma 2, we need to show that the control gain and value learned by the PI Algorithm 1 at each iteration are close enough to that learned by model-based PI algorithms. We will prove this lemma by induction.

- 1) When the iteration $k = 0$, we automatically have $K_i^{(0)} = \bar{K}_i^{(0)}$ for all the followers i . Let $\Delta \bar{P}_i^{(0)} = P_i^{(0)} - \bar{P}_i^{(0)}$. It is checkable that $\Delta \bar{P}_i^{(0)}$ is a least-squares solution to the following equation:

$$\Psi_{\text{PI},i}^{(0)} \begin{bmatrix} \text{vecs}(\Delta \bar{P}_i^{(0)}) \\ \text{vec}(R_i^{-1} \bar{B}_i^T \Delta \bar{P}_i^{(0)}) \\ \text{vec}(\Theta_{i0}^T \Delta \bar{P}_i^{(0)}) \\ \text{vec}(\Xi(\Delta \bar{P}_i^{(0)})) \end{bmatrix} = 2\Gamma_{\xi_i, \rho_i} \text{vec}(\bar{P}_i^{(0)})$$

where

$$\Xi(P) = [P\Theta_{i1}, P\Theta_{i2}, \dots, P_i^{(k)}\Theta_{i j_{iN+1}}]^T.$$

Let $\|\rho\| := \max_{t_0 \leq t \leq t_s} \rho(t)$. Then, we can obtain that $\lim_{\|\rho\| \rightarrow 0} \bar{P}_i^{(0)} - P_i^{(0)} = 0$, and $\lim_{\|\rho\| \rightarrow 0} \bar{K}_i^{(1)} - K_i^{(1)} = R_i^{-1} \bar{B}_i^T (\bar{P}_i^{(0)} - P_i^{(0)}) = 0$.

- 2) When the iteration $k = j$, suppose that we have $\lim_{\|\rho\| \rightarrow 0} \bar{K}_j^{(0)} - K_j^{(0)} = 0$. Let $\Delta \bar{P}_i^{(j)} = P_i^{(j)} - \bar{P}_i^{(j)}$. It is checkable that $\Delta \bar{P}_i^{(j)}$ is a least-squares solution to the following equation:

$$\Psi_{\text{PI},i}^{(j)} \begin{bmatrix} \text{vecs}(\Delta \bar{P}_i^{(j)}) \\ \text{vec}(R_i^{-1} \bar{B}_i^T \Delta \bar{P}_i^{(j)}) \\ \text{vec}(\Theta_{i0}^T \Delta \bar{P}_i^{(j)}) \\ \text{vec}(\Xi(\Delta \bar{P}_i^{(j)})) \end{bmatrix} = \Delta \bar{\Phi}_i^{(j)}$$

where $\lim_{\|\rho\| \rightarrow 0} \Delta \bar{\Phi}_i^{(j)} = 0$. Then, we can obtain that $\lim_{\|\rho\| \rightarrow 0} \bar{P}_i^{(j)} - P_i^{(j)} = 0$, and $\lim_{\|\rho\| \rightarrow 0} \bar{K}_i^{(j+1)} - K_i^{(j+1)} = R_i^{-1} \bar{B}_i^T (\bar{P}_i^{(j)} - P_i^{(j)}) = 0$.

Increasing μ_1 and μ_2 can accelerate the convergence of ρ . In other words, one can always find large enough μ_1 and μ_2

(small enough $\|\rho\|$) such that the statement of Lemma 2 holds. The proof is thus completed.

APPENDIX B PROOF OF LEMMA 3

In order to complete the proof of Lemma 3, we need to show that the control gain and value learned by the VI Algorithm 2 at each iteration are close enough to that learned by model-based VI algorithms. Similar to the proof of Lemma 2, we will prove Lemma 3 by induction.

- 1) When the iteration $k = 0$, we automatically have $P_i^{(0)} = \hat{P}_i^{(0)}$ and $K_i^{(0)} = \hat{K}_i^{(0)}$.
- 2) When the iteration $k = j$, suppose that we have $\lim_{\|\rho\| \rightarrow 0} \hat{P}_j^{(j)} - P_j^{(j)} = 0$ and $\lim_{\|\rho\| \rightarrow 0} \hat{K}_j^{(j)} - K_j^{(j)} = 0$. Let $\Delta \hat{P}_i^{(j)} = P_i^{(j)} - \hat{P}_i^{(j)}$. It is checkable that the following equation holds:

$$\Psi_{\text{VI},i} \begin{bmatrix} \text{vecs}(\Delta \mathcal{H}_i^{(j)}) \\ \text{vec}(R_i^{-1} \bar{B}_i^T \Delta \hat{P}_i^{(j)}) \\ \text{vec}(\Theta_{i0}^T \Delta \hat{P}_i^{(j)}) \\ \text{vec}(\Xi(\Delta \hat{P}_i^{(j)})) \end{bmatrix} = \Delta \hat{\Phi}_i^{(j)}$$

where $\Delta \mathcal{H}_i^{(j)} = \bar{A}_i^T \Delta \hat{P}_i^{(j)} + \Delta \hat{P}_i^{(j)} \bar{A}_i$ and $\lim_{\|\rho\| \rightarrow 0} \Delta \hat{\Phi}_i^{(j)} = 0$. Then, we can obtain that $\lim_{\|\rho\| \rightarrow 0} \hat{P}_i^{(j+1)} - P_i^{(j+1)} = 0$, and $\lim_{\|\rho\| \rightarrow 0} \hat{K}_i^{(j+1)} - K_i^{(j+1)} = R_i^{-1} \bar{B}_i^T (\hat{P}_i^{(j+1)} - P_i^{(j+1)}) = 0$.

Therefore, one can always find large enough μ_1 and μ_2 (small enough $\|\rho\|$) such that the statement of Lemma 3 holds. The proof is thus completed.

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